

Draft - expert review pending

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Topic 1.2 - People, activities and environmental problems

A real classroom situation

In a boys' high school in lower Sindh, the science teacher takes his Class 9 pupils to the edge of a field near the school. The crop stands in patches: some plants green, some yellow and dying. An old farmer tells them, "Our fathers grew good wheat here. Now the water stands too long in some parts and the soil has turned white in others." Back in the classroom, the teacher writes two words on the board: waterlogging and salinity. "This field," he says, "is not just a farming problem. It is a story about us."

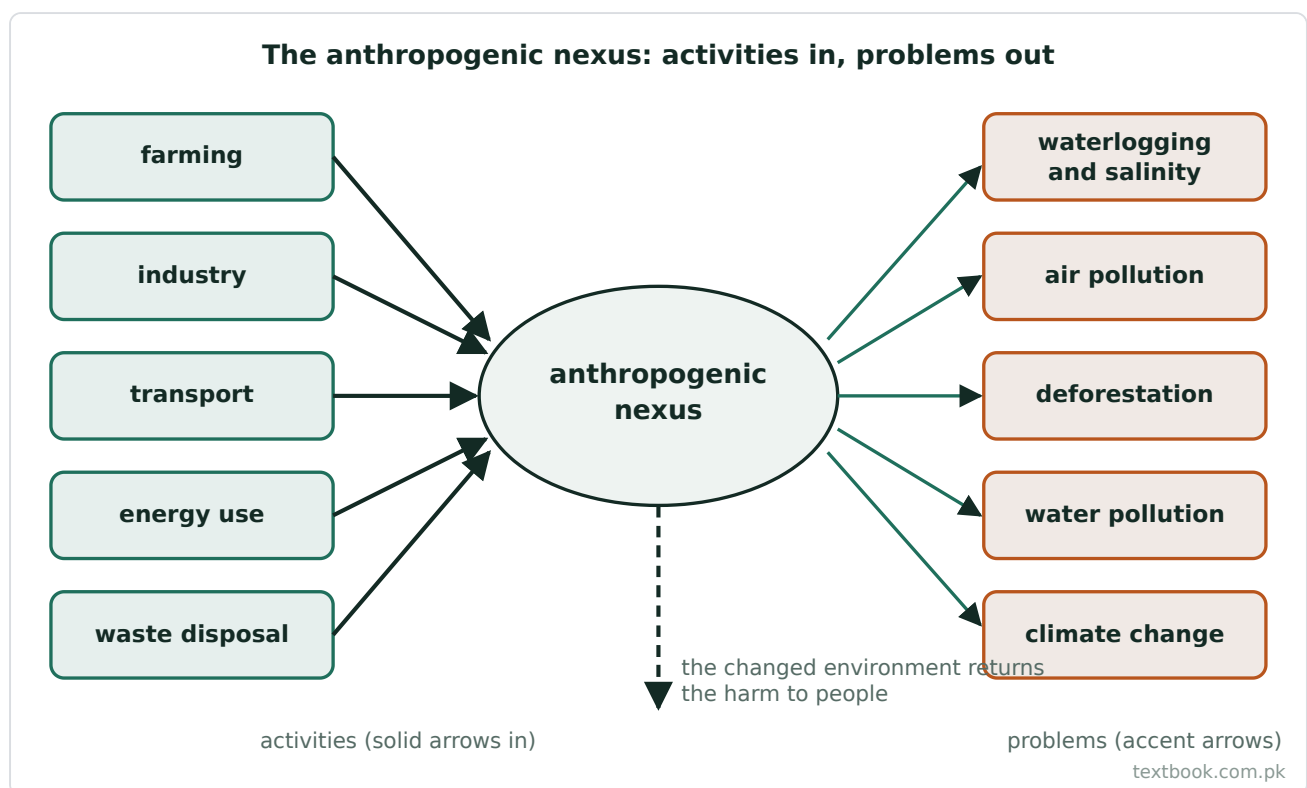
The field is a story about people changing their environment, and the environment answering back. This topic is about that two-way relationship, which the course guide calls the nexus between environmental problems and anthropogenic activities, and about how such problems appear at three scales: global, regional and local.

Explanation

The nexus between environmental problems and human activities

Anthropogenic means caused by people. An anthropogenic environmental problem is one that either did not exist before human activity created it, or existed only mildly and was made worse by it. The nexus is the two-way link: activities change the environment, and the changed environment returns the harm to people, sometimes to the same people, often to different and poorer people, and sometimes years later.

The chain usually runs in four steps. An activity, such as farming, industry, transport, energy use or waste disposal, takes something from the environment or puts something into it. The environment responds: water tables rise, air goes grey, forests thin. The response crosses back into human life: crops fail, illness rises, floods damage homes. And the human response to that harm often starts the chain again, as when a failing crop pushes farmers to clear more land.



A worked example, fully traced. Take the yellow field near the Sindh school. Step one, the activity: farmers in the last century expanded canal irrigation across lower Sindh to grow more food. Step two, the environmental response: canals seeped, fields were watered more than the soil could drain, the water table rose, and the salts naturally present in the soil were drawn up and left on the surface as the water evaporated. Step three, the harm returned: waterlogged and salt-affected soil yields less, and the white crust you can see is

the evidence. Step four, the response: some farmers abandon the land or dig tube wells, which can draw up even saltier water. One chain, four steps, every link human and environmental at once.

The same four-step structure fits the other big activities. Industry and transport burn fuel and load the air with pollutants. Energy choices, from coal to firewood, drive both local smoke and global greenhouse gases. Waste disposal, when it means open dumping or burning, moves the problem from the bin to the air, the soil and the water. The United Nations Decade on Ecosystem Restoration describes the present situation plainly: there has never been a more urgent need to restore damaged ecosystems (UN, 2021).

Global, regional and local issues and their current status

Environmental problems are usually sorted by how far their causes and effects reach.

- **Global issues** affect the whole planet, whatever their source. Climate change is the clearest case: greenhouse gases mix through one shared atmosphere, and the World Health Organization reports that about 3.6 billion people already live in places highly susceptible to its effects (WHO, 2023). The loss of biodiversity is the second: the WHO also reports that around one million species face extinction risk (WHO, 2025). Ozone depletion, now slowly recovering thanks to a global agreement, is the third.
- **Regional issues** cross borders or cover a region. South Asia shares the smog that builds each winter, the waters of the Indus and its tributaries, and the Himalayan glaciers whose melt feeds the rivers. The Indus system's stress is a regional issue: what happens upstream in one country changes the flow downstream in another.
- **Local issues** are tied to a place, though they repeat everywhere. A neighbourhood's burning garbage heap, a village's contaminated hand pump, a school's dusty courtyard, one city's traffic noise.

Environmental issues at three scales

Scale	Example issue	Current status (one line)	Who feels it
Global	climate change	3.6 billion people already live in highly susceptible areas (WHO, 2023)	everyone; the poorest first and worst
Regional	Indus water stress	withdrawals already exceed a highly stressed threshold; population adds pressure	all basin countries; farmers and delta communities hardest
Local	neighbourhood waste burning	repeated across cities; smoke enters homes and schools nearby	the street's own residents, especially children and elders

The scales connect: a local heap adds carbon to the global atmosphere; a global rise in temperature shows up locally as a Karachi heatwave.

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The scales are connected, which is the point of showing them together. A local burning heap adds carbon to the global atmosphere. A global rise in temperature shows up locally as heatwaves in Karachi. When you analyse an issue, name its scale, but also ask how it crosses to the others.

Common misconception. Many people think global problems are the "real" environmental problems and local ones are too small to count. The reverse error also appears: that only local problems matter because they are the ones we can see. Both are wrong. Global problems are built from millions of local activities, and local people suffer global problems first and worst. The scales are one system, not a competition.

Activity: Headline scale sort

Work in groups of three. Your teacher will give you one week of environmental headlines from a Pakistani newspaper. Sort each headline into global, regional or local. For two headlines, one that surprised your group and one you argued about, write one sentence naming the human activity behind the problem and one sentence naming a connection to a different scale. Allow 25 minutes, then each group presents its two headlines.

Check your understanding

1. Define "anthropogenic" in one sentence and give one anthropogenic problem from your own district. (*Remember*)
2. Trace the four steps of the nexus for one activity other than irrigation, such as transport or waste burning. (*Apply*)
3. A classmate says, "Smog in Lahore is a local issue." Give two reasons why the regional and global scales are also involved. (*Analyze*)
4. Name one current global environmental issue and state one figure from this topic that describes its status. (*Remember*)

Summary

Environmental problems and human activities are linked in a two-way nexus: activities change the environment, and the changed environment returns harm to people. The chain runs activity, environmental response, harm returning, human response. Irrigation-induced waterlogging and salinity in lower Sindh is one complete worked chain. Issues appear at three scales, global such as climate change and biodiversity loss, regional such as the Indus system and winter smog, and local such as a burning waste heap, and the scales connect in both directions.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define anthropogenic and use it correctly in a sentence.
- ☐ I can trace a human activity through the four nexus steps to its returning harm.
- ☐ I can classify an environmental issue as global, regional or local with a reason.
- ☐ I can explain how one local issue connects to a global one, with an example.

Try this at your practicum school

Find the nearest environmental problem to the school gate, a drain, a heap, a dusty road, a smoky stove. In half a page, name the scale you would call it, the human activity that drives it, and one way it connects to a bigger scale. Ask one person who lives nearby how long it has been there. Do not name the school or the person.

Summative task

Write 150 to 200 words on the following.

Choose ONE local environmental issue you have personally seen in Sindh. Trace it through the four nexus steps, then explain one connection it has to a regional or global issue. Finish with one sentence naming who bears most of its harm.

Mini-rubric

Criterion	What a strong response does
Nexus traced (Analyze)	Names all four steps in order, with the specific activity and returning harm
Scale judged (Understand)	Calls the issue local correctly and names the cross-scale connection precisely
Evidence (Apply)	Uses a real, specific example from Sindh, not a generic "pollution"
Clarity and register	Plain sentences; within the word count

Further reading

- World Health Organization. (2023). *Climate change* (fact sheet). Global status and human susceptibility: <https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health>

- World Health Organization. (2025). *Biodiversity* (fact sheet). The global species-extinction status: <https://www.who.int/news-room/fact-sheets/detail/biodiversity-and-health>
- Haines, A., & Frumkin, H. (Eds.). *Planetary Health*. Island Press. (guide-recommended book, cited by reference only)

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